

MOHAMMAD YAKUB

📍 Hyderabad, Telangana, India

✉️ [gmail.com](#)  [LinkedIn](#)  [GitHub](#)

SUMMARY

AI/ML Engineer with hands-on experience building **Machine Learning, Deep Learning, Generative AI, and RAG systems**. Skilled in Python and experienced in developing end-to-end AI pipelines, LLM evaluation workflows, hybrid search systems, and computer vision applications. Strong foundation in **Data Structures & Algorithms, Object-Oriented Programming, and Software Engineering**.

EXPERIENCE

Research Intern

May 2024 – Jul 2024

Humanoid Lab, IIIT Dharwad

Contributed to research and development of AI-based speech processing solutions by designing, implementing, and evaluating machine learning pipelines in a collaborative research environment.

Developed an automated audio data pipeline using **yt-dlp** and **pydub**, trained a **Retrieval-based Voice Conversion (RVC)** model, and achieved an **F0 RMSE of 39.54 Hz**.

EDUCATION

B.Tech in Artificial Intelligence & Data Science, KL University

2022 – 2026

CGPA: **9.26**

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Software Engineering, Cloud Computing, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

SKILLS

Languages: Python, C++, Java, SQL, JavaScript, TypeScript

Core: Data Structures Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Generative AI, RAG, Software Engineering

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, LLM Evaluation, Prompt Engineering

Frameworks/Tools: FastAPI, Git, GitHub Actions, Docker, VS Code, Google Colab, n8n

Databases: MySQL, PostgreSQL, SQLite, MongoDB, ChromaDB

PROJECTS

Model Regression Detection System

August 2026 – September 2026

*Tech Stack: **Python, OpenAI, Groq, Pydantic, GitHub Actions, Docker***

[GitHub](#)

- Developed a CI/CD-style LLM evaluation pipeline that tests prompt changes against a **72-case golden dataset** to detect accuracy regressions and model drift.
- Implemented **baseline comparison, LLM-as-a-Judge evaluation, drift detection, HTML reporting, and Slack alerts** to identify and communicate model performance degradation.
- Integrated **GitHub Actions and Docker** to automatically evaluate prompt changes and block merges when critical regressions exceed configured thresholds.

RAG Pipeline with Hybrid Search

August 2026 – September 2026

*Tech Stack: **Python, FastAPI, ChromaDB, BM25, Cross-Encoder, RRF***

[GitHub](#)

- Developed a production-grade RAG pipeline combining **dense vector search and BM25 sparse retrieval** with Reciprocal Rank Fusion (RRF) for improved document retrieval.
- Implemented **cross-encoder reranking, grounded response generation, verified citations, and confidence gating** to improve answer relevance and reliability.
- Built a **52-question golden-set evaluation harness** with multiple retrieval and answer-quality metrics to systematically evaluate pipeline performance.

Automatic Braking System for Forward and Backward Collisions

March 2025 – July 2025

*Tech Stack: **Python, TensorFlow, Keras, Random Forest, OpenCV***

Developed an AI-powered Automatic Braking System to detect forward and backward vehicle collisions using sensor data and computer vision techniques.

- Designed and evaluated Random Forest and Neural Network models, achieving **75% accuracy** in collision prediction.
- Built an end-to-end pipeline for data preprocessing, model training, evaluation, and performance visualization using confusion matrices and ROC curves.

LEADERSHIP EXPERIENCE

Editing Lead

2023 – 2026

Synapse Tech Club, KL University

- Led a **5-member** content team, streamlining editorial workflows and increasing student engagement by **40%** through a revamped weekly newsletter reaching over **800** students.